Premium of a call with K = 100 USD/share (9 tree steps)

Leisen–Reimer tree: a smooth curve
CRR tree: straight pieces joined at corners
one CRR piece, extended
corner: a terminal node crosses the strike

open circles: $S_0 - \epsilon$, S_0, $S_0 + \epsilon$ lie on one line,
so the central difference gives Gamma = 0

Premium (USD/share)

12.5
10.0
7.5
5.0
2.5

90
95
100
105
110

Delta: the slope of the premium

Leisen–Reimer tree: changes smoothly
CRR tree: flat on each piece, jumps at each corner

Delta (share/share)

1.0
0.8
0.6
0.4
0.2
0.0

90
95
100
105
110

Share price today S_0 (USD/share)